A Bi-Objective Programming Model for Multimodal Transportation Routing Problem of Bulk Cargo Transportation

Apichit Maneengam
Department of Mechanical Engineering Technology, College of Industrial Technology
King Mongkut's University of Technology North Bangkok
Bangkok, Thailand
e-mail: apichit.a@cit.kmutnb.ac.th

***Abstract*—In this paper, we propose a bi-objective programming model to select the optimal route for moving a consignment of goods between the anchorage area at the Gulf of Thailand to factories throughout Thailand using a multi-modal transportation network and joint consideration of transportation mode selection. The bi-objective programming model is constructed with features based on two objective functions - minimizing transportation costs and minimizing CO_2 emissions, subject to transportation jobs that should be completed before their respective due dates. The solutions are obtained by the ε-constraint method, after which a Pareto frontier will be plotted. The decision makers can make a decision tradeoff between the two objectives from that plot.**

Keywords-component; multimodal transportation routing problem; bi-objective programming model; ε-constraint method

I. INTRODUCTION

Following trade globalization, the conventional road mode is no longer a sustainable, feasible solution, necessitating other means of transportation [1]. Multimodal transportation is a key supply chain component to ensure the efficient movement and timely availability of raw materials and finished products [2]. Where one operator is responsible for transport activities from the shipper's door to the consignee's door, a multimodal transport operator (MTO) acts as a "carrier" with the responsibility and liability to perform transport activities for the entire transport process.

It is important for an MTO to have the ability to design and provide effective transport arrangements, ensuring the continuous movement of assigned goods along the best possible route and by the most efficient means in order to meet each customer's delivery requirements, this problem is called multimodal transport problem. The multimodal transport problem (MTP) is the decision of transportation route from the origin to the destination. In the MTP the shortest path problems need to be solved under additional constraints or on particular structured graphs. Every node stands for a terminal or sub-destination that is on the transportation route. Each mode of transport has corresponding transportation means and infrastructures. Thus, the study of multimodal transport problems is important in both its theoretical and practical significance. Moreover, MTOs usually need more Pareto frontier solutions so they can choose the most desirable solution according to their particular requirements.

Currently, the dry bulk freight volume of all transportation modes has increased rapidly. Bulk cargo transportation is often described by the full truckload transport problem and product cargo that is transported in large quantities by large trucks, trains or ocean liners [3]. To move the bulk cargo from an original to a destination in Thailand the multimodal transportation is the main issue.

In this paper, researcher deal with a real-world problem and focus particularly on the bulk-cargo multimodal transport problem. A bi-objective programming model is applied to select the best transportation routes and mode of transportation subject to due date constraints. Minimizing the total transportation costs, the total transportation time and total CO_2 emissions are set as the optimization objectives. The solutions are obtained by ε-constraint method [4], then a Pareto frontier is introduced. The remainder of this paper is organized as follows. Section 2 proposes a review of literature. A bi-objective programming model is formulated in Section 3. Section 4 the ε-constrained method is described. Section 5 presents the transportation network of the case study, while the results are illustrated in Section 6. Finally, the conclusions are given in Section 7.

II. LITERATURE REVIEW

Research in the area of multimodal transportation planning has accelerated in the last decade, urging for an update. Manee-ngam et al. [5] presented dynamic programming to select multimodal transportation routes in order to minimize the total cost of transport activities in Thailand. Xie et al. [6] formulated mixed-integer liner programming for a multimodal hazardous materials problem in order to minimize total risk and total cost. Xiong and Wang [7] proposed a bi-level multi-objective Taguchi genetic algorithm for a multimodal routing problem with time windows to minimize the total transportation cost and total transportation time. Sun and Lang [8] proposed a bi-objective optimization model to rectify the multi-modal transportation routing planning problem. Minimizing the total transportation cost and total transportation time are set as the optimization objectives of the model. Rabbani et al. [9] formulated two stages mathematically for a multimodal transportation system, in which waste packs are transferred to the final destination, while travel time and the risk of environmental threats are minimized. Abbassi et al. [10] present a bi-objective intermodal transportation problem to minimize both the transportation cost and maximize

978-1-7281-6785-5/20/$31.00 ©2020 IEEE

Authorized licensed use limited to: UNIVERSITY OF TENNESSEE LIBRARIES. Downloaded on March 20,2024 at 15:26:39 UTC from IEEE Xplore. Restrictions apply.

overtime to delivery products. They proposed a non-dominated sorting genetic algorithm improved by a local search heuristic and the greedy randomized adaptive search procedure with iterated local search heuristics are applied to the problem. Recently, Zan et al. [11] proposed new Pareto-based genetic algorithm to solve multi-objective scheduling problems of automated manufacturing systems.

The literature review reveals that the usual approaches to selection of multimodal transportation routes have emphasized on selecting multimodal transportation route in order to minimize cost or minimize time without considering the significance of the greenhouse gas (GHG) setting, which is important to a number of countries' attention in the model. In the future, proactive GHG management will be essential for the transportation industry as well as for organizations managing multimodal transportation hubs that integrate several types of freight carriers, logistics services and manufacturing or processing activities at a single site [12]. Thus, this study focuses on two objectives: minimizing the total transportation cost and total CO_2 emissions.

III. Modelling of Multimodal Routing Problem

This section describes the model of the multimodal routing problem with due date constraints for the design of optimal strategy in the following section. Let G(A, J, M) denote the multimodal transportation network, where A, J and M are a set of nodes, a set of links and a set of transportation modes, respectively. Let o and d denote the origin node and destination node, respectively. The parameters and decision variables in the model are defined as follows.

Parameter

c_{ij}^m Transportation cost on link (i, j) by transportation mode m. (i, j) ∈ J, i ∈ A, j ∈ A, m ∈ M

d_{ij}^m Distance on link (i, j) by transportation mode m.

h_j^m Cost at node j by transportation mode m.

F^m The number of vehicle by transportation mode m

P_j^m The number of berth at node j by transportation mode m

s_j^m Transshipping time at node j by transportation mode m

Q Quantity of task

D Due date of task

Cap^m Vehicle capacity of each transportation mode m

e_{em}^m Emission factor of fuel consumption for empty loading transportation mode m

e_{full}^m Emission factor of full loading fuel consumption

e_j^w Electricity emission factor at node j

e_j^γ Fuel consumption emission factor to store goods at node j

w_j Electricity consumption to store goods at node j

γ_j Fuel consumption for store goods at node j

Variable

g_{ij}^m The waiting time of node i, while the vehicle is transporting cargo from node i to j by transportation mode m.

T_{ij}^m Transportation time on link (i, j) by transportation mode m

α_{ij}^m CO2 emission of transportation on link (i, j) by transportation mode m.

β_j^m CO2 emission of transshipping at a node j by transportation mode m.

t_{ij}^m Travel time on link (i, j) by mode of transport m.

Decision variable

x_{ij}^m Flow of commodity on link (i, j) by transportation mode m.

y_j^m Binary decision variable. 1 = If the consignment of goods is transshipped by transportation mode m at node j. 0 = otherwise

The bi-objective optimization for multimodal routing problem can be formulated as follows.

$$Min\ Z_1 = \sum_{(i,j)\in J}\sum_{m\in M} c_{ij}^m x_{ij}^m + \sum_{i\in J}\sum_{m\in M} h_j^m y_j^m \tag{1}$$

$$Min\ Z_2 = \sum_{(i,j)\in J}\sum_{m\in M} \alpha_{ij}^m x_{ij}^m + \sum_{i\in J}\sum_{m\in M} \beta_j^m y_j^m \tag{2}$$

Subject to

$$\sum_{(i,j)\in J} x_{ij}^m - \sum_{(j,i)\in J} x_{ji}^m = \begin{cases} 1, & i = o \\ -1, & i = d \\ 0, & otherwise. \end{cases} \quad \forall i \in J, m \in M \tag{3}$$

$$\sum_{(i,j)\in J}\sum_{m\in M} T_{ij}^m x_{ij}^m y_j^m \le D \tag{4}$$

$$\sum_{i\in I} x_{ij}^m - y_j^m = 0 \quad \forall i \in A, m \in M \tag{5}$$

$$\sum_{m\in M} x_{ij}^m \le 1 \quad \forall (i,j) \in J \tag{6}$$

$$x_{ij}^m \in \mathbb{Z}_+ \quad \forall (i,j) \in J, m \in M \tag{7}$$

$$y_j^m \in \{0.1\} \quad \forall (i,j) \in J, m \in M \tag{8}$$

Where

$$T_{ij}^m = \sum_{(i,j)\in J}\sum_{m\in M} \left\{ \left[\left(\left\lceil \frac{F^m}{P_j^m} \right\rceil s_i^m \right) \cdot \left(\frac{Q}{F^m Cap^m} \right) \right] + G_{ij}^m \right\} \tag{9}$$

$$G_{ij}^m = \begin{cases} g_{ij}^m, & \left[\left(t_{ij}^m + t_{ji}^m + s_j^m \right) > \left\lceil \frac{F^m}{P_i^m} \right\rceil s_i^m \right] \wedge \left[\left(\frac{Q}{F^m Cap^m} \right) > 1 \right] \\ t_{ij}^m, & \left[\left(\frac{Q}{F^m Cap^m} \right) = 1 \right] \\ 0, & otherwise \end{cases} \tag{10}$$

$$g_{ij}^m = \left[\left(t_{ij}^m + t_{ji}^m + s_j^m \right) - \left\lceil \frac{F^m}{P_j^m} \right\rceil s_i^m \right] \cdot \left[\left(\frac{Q}{F^m Cap^m} \right) - 1 \right] \tag{11}$$

$$\alpha_{ij}^m = \sum_{m\in M} \left[\left(e_{ij}^{m,em} d_{ij}^m + e_{ij}^{m,full} d_{ij}^m \right) \left(\frac{Q}{F^m Cap^m} \right) \right] \tag{12}$$

Authorized licensed use limited to: UNIVERSITY OF TENNESSEE LIBRARIES. Downloaded on March 20,2024 at 15:26:39 UTC from IEEE Xplore. Restrictions apply.

$$\beta_i^m = e_j^w w_j + e_j^\gamma \gamma_j \quad (13)$$

In this formulation, the first objective function is to minimize the total transportation cost as (1). In (2), the third objective is reduction of total CO2 emissions for the consignment of goods. The constraint Eq.(3) is the network flow equilibrium constraint for each node [7], [8] Constraint (4) means the total transportation time should not exceed the due date of goods. Constraint (5) indicates the relationship between the two decision variables; if there is trans-shipping at a node, two arcs linking it and its conjoint nodes must be covered in the transportation route. Otherwise, these two arcs should not be covered. Constraint (6) ensures that the consignment of goods will not be transported by splitting into several sub consignments. The constraint sets (7)–(8) represent the integrality conditions on the decision variables. Equations (9)-(11) explain the details of the estimated transit time from node i to j by transportation mode m. Equation (12) is a calculation of CO_2 emissions involving the estimation of the amount of CO_2 released from energy used in transportation during the distance between the origin and destination locations. Equation (13) is a calculation of CO_2 emissions released from energy used in trans-shipping at any terminal. In addition, some paths cannot be transported through, such in case the routes are being repaired or are flooded. The Big M method is used to determine the value of M for the cost of those paths.

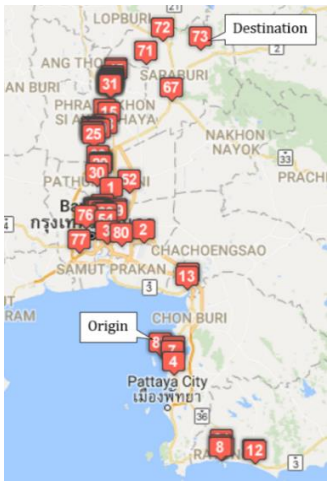


Figure 1. Locations of the terminals under study

IV. A Case Study

The case study is a transportation service provider of bulk cargo (mainly coal) in Thailand. The company is responsible for transporting bulk goods from the anchorage area in the Gulf of Thailand (node number 83) to customers inland (node number 73). The transportation networks are depicted in figure 1. These instances and all parameters can be downloaded from https://sites.google.com/site/transportationlibrary. In this case, there are 83 nodes and 9,579 links, which are shown in figure 2. There are three maximum available transportation modes corresponding to dump truck, barge or lighter and vessel between any two nodes. Each terminal has 5 berths worked in parallel. The transport job having 20,000 tons in capacity of each job is equal to the capacity of a vessel. The MTO and all carriers should be completed this job before 100 working hours. The transportation mode is plotted using different type of line.

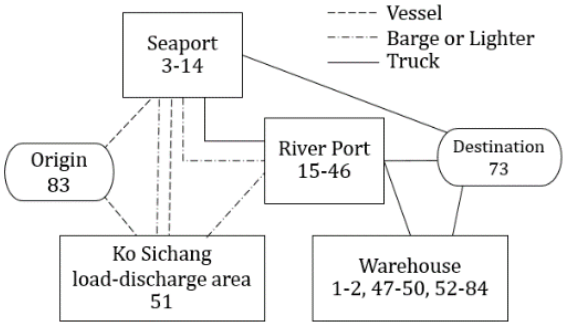


Figure 2. The 83-node multimodal transportation network.

Different transportation routes have different distances between any two nodes. In addition, we calculated the CO_2 emissions from the emission factors of transportation types via referencing the method of estimation and emission factors [13], [14] following the Thailand Greenhouse Gas Management Organization (TGO) criteria.

In this case study, the MTO (Multimodal Transport Operator) must generate a transportation route that includes where to load and unload, and designate the means of conveyance. At present, MTO use the method proposed by Manee-ngam et al. [5] to select multimodal transportation routes in order to minimize the total cost, without other objective functions. However, they also have a policy to reduce future CO2 emissions. Therefore, they need a simple method (in both implementation and use) to solve the bi-objective multi-modal transportation routing planning problem. This must be an effective method to generate the Pareto front because the MTO can select a suitable Pareto solution as the most convenient transportation system, according to the Pareto frontier [8]. The needs of the MTO will be examined in the next section to choose the most appropriate method.

V. Multi-objective Optimization Techniques

There are many techniques that appear in the literature for solving multi-objective problems within the Operations Research community, include: linear combination of weights method [15], global criterion method [16], ε-constraint method [4], and multi-objective genetic algorithm [17].

In this paper, we have compiled literature which compares these techniques for use as guidelines when choosing the means to solve the problems described in this case study. Mavrotas [18] explains the advantages and advantages of the ε-constraint and weighting method and found that the former has several advantages. This is because it is simpler and can control the quantity of generated

Authorized licensed use limited to: UNIVERSITY OF TENNESSEE LIBRARIES. Downloaded on March 20,2024 at 15:26:39 UTC from IEEE Xplore. Restrictions apply.

efficient solutions by properly adjusting the number of grid points in each of the objective function ranges. Later, Chiandussi, et al. [19] analyzed and compared four techniques for solving the multi-objective optimization problem by describing their formulation, advantages, and disadvantages. They explained that the ε-constraint method allows for an effective reduction of the computational cost. This is particularly convenient when the objective function bounds are known, and it is particularly efficient with convex and linear Pareto fronts. The ε-constraint method is always able to identify the best compromise solutions belonging to the Pareto front, independently of its shape. Consequently, the ε-constraint method showed the best results. Recently, Demir et al. [20] presented a bi-objective linear mixed-integer mathematical formulation of the green intermodal service network and compared the following three techniques to resolve this problem: the basic weighting method, the weighting method with normalization and the ε-constraint method. From a comparison of the different bi-objective optimization approaches shown, the solutions provided by the ε-constraint method represent a larger area than the other methods, and also obtained more Pareto solutions because it is more applicable for both objectives.

From the literature review, all research studies recommend using the ε-constraint method because it is simpler in both implementation and use, and is a more effective method for generating the Pareto front than other methods. The above advantages are also in line with the needs of the MTO. Therefore, the Pareto frontier is gained by using the ε-constraint method proposed by Chankong and Haimes [4], which include the following steps.

First, the optimal values of the two objectives denoted by $f_1(x)$ and $f_2(x)$ can be gained by solving the single objective optimization model with (1), (3)-(13) and (2), (3)-13) respectively.

Second, optimize one of the objective function j from two objectives as shown in (14), the other objective functions are transformed in constraints fixing threshold values of the constrained objective functions k (ε_k) over them, with $k \in p$, $k \neq j$ as shown in (15). Therefore the problem is substituted by the ε-constraint problem:

$$\min_{x \in X} f_j(x) \tag{14}$$

Subject to:

$$f_k(x) \leq \varepsilon_k \quad k = 1, \dots, p, k \neq j \tag{15}$$

Where x is the vector of decision variables are the p objective functions. By parametrical variation in the RHS of the constrained objective functions (ε_k) the efficient solutions of the problem are obtained.

VI. Result

Gurobi 5.5 was used to solve the bi-objective optimization model based on ε-constraint method [3] for multimodal routing problem directly with the default algorithm setting. The proposed method is tested on transportation requests from the anchorage area in the Gulf of Thailand to customers' factories in Thailand. Researcher get 20 results, but most of them have duplicate solutions. Thus, we have presented only unique results or those with no duplicate results, as shown in Table I. The Pareto frontier or tradeoff curve will be plotted, as shown in figure 3. The MTO can select one of the Pareto solutions as the transportation scheme conveniently based on the Pareto frontier, the MTO's preference to a specific objective and the evaluation of the customer's time satisfaction. For example, if the customer is satisfied that the total CO_2 emissions should not exceed 280,000 CO_2e for this task, then the MTO can plan the transportation scheme by using the 5th Pareto solution. In addition, the Pareto frontier clearly indicates that an increase in the total transportation cost will lead to a decrease in CO_2 emissions.

TABLE I. The Pareto Optimal Solution

NO.	Total cost (Baht/ton)	CO_2 emissions (CO_2e)	Optimal path
1	586	223,876	83+3>73
2	586	230,796	83+13-46>73
3	541	230,795	83+14-46>73
4	344	275,956	83+51-17>73
5	344	285,871	83+51-46>73

+ Vessel mode, - Barge or Lighter mode, > Truck mode
*,# is the best values of Z_1, Z_2, respectively

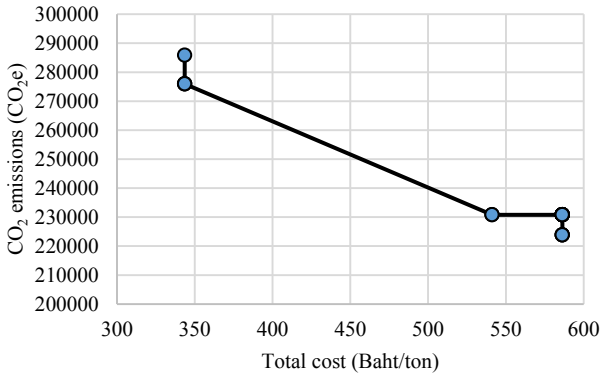


Figure 3. The Pareto frontier of the bi-objective optimization.

VII. Conclusions

In this paper, a bi-objective optimization model is proposed for multimodal routing problems within the context of the case study. The solutions are obtained by ε-constraint method [4], after which decision makers can use the Pareto frontier to make decisions and modify strategies when planning the multi-modal transportation organization. The results show that the model can effectively describe the multimodal transportation network and the Pareto optimal front, which provides a series of transportation paths and transportation modes. Multi-job integration is temporarily excluded in this study. In the future, we will try to address a novel version of the ε-constraint method to improve its performance, or adapt it to a specific type of problem. Furthermore, a risk analysis will be performed.

Authorized licensed use limited to: UNIVERSITY OF TENNESSEE LIBRARIES. Downloaded on March 20,2024 at 15:26:39 UTC from IEEE Xplore. Restrictions apply.

ACKNOWLEDGMENT

This research is partially supported by the College of Industrial Technology, King Mongkut's University of Technology North Bangkok.

REFERENCES

[1] M. Steadieseifi, N. P. Dellaert, W. Nuijten, T. Van Woensel, and R. Raoufi, "Multimodal freight transportation planning: A literature review," European Journal of Operational Research, vol. 233, no. 1, pp. 1–15, 2014, doi:10.1016/j.ejor.2013.06.055.

[2] T. Crainic, "International Series in Operations Research & Management Science," in Handbook of Transportation Science, 2nd ed., R. Hall, Ed. New York: Springer US, 2003, pp. 451–516, doi:10.1007/b101877.

[3] A. Maneengam and A. Udomsakdigool, "Solving the collaborative bidirectional multi-period vehicle routing problems under a profit-sharing agreement using a covering model," International Journal of Industrial Engineering Computations, vol. 11, no. 2, pp. 185–200, 2020, doi:10.5267/j.ijiec.2019.10.002.

[4] V. Chankong and Y. Haimes, Y ., Multiobjective Decision Making: Theory and Methodology. Amsterdam: North-Holland, 1983.

[5] A. Manee-ngam, W. Laotaweesub, A. Udomsakdigool, and K. Sripathomswat, "Applying Dynamic Programming for Solving the Multimodal Transport Problem : A Case Study of Thai Multimodal Transport Operator," in Asia Pacific Industrial Engineering & Management Systems Conference 2012, 2012, pp. 1319–1325.

[6] Y. Xie, W. Lu, W. Wang, and L. Quadrifoglio, "A multimodal location and routing model for hazardous materials transportation," Journal of Hazardous Materials, vol. 227–228, pp. 135–141, 2012, doi: 10.1016/j.jhazmat.2012.05.028.

[7] G. Xiong and Y. Wang, "Best routes selection in multimodal networks using multi-objective genetic algorithm," Journal of Combinatorial Optimization, vol. 28, no. 3, pp. 655–673, 2014, doi:10.1007/s10878-012-9574-8.

[8] Y. Sun and M. Lang, "Bi-objective optimization for multi-modal transportation routing planning problem based on pareto optimality," Journal of Industrial Engineering and Management, vol. 8, no. 4, pp. 1195–1217, 2015, doi:10.3926/jiem.1562.

[9] M. Rabbani, S. Sadri, and H. Rafiei, "A multimodal transportation system routing implemented in waste collection," Decision Science Letters, vol. 5, pp. 61–80, 2016, doi:10.5267/j.dsl.2015.8.003.

[10] A. Abbassi, A. Elhilali, and J. Boukachour, "Modelling and solving a bi-objective intermodal transport problem of agricultural products," International Journal of Industrial Engineering Computations, vol. 9, pp. 439–460, 2018, doi:10.5267/j.ijiec.2017.12.001.

[11] X. Zan, Z. Wu, C. Guo, and Z. Yu, "A Pareto-based genetic algorithm for multi-objective scheduling of automated manufacturing systems," Advances in Mechanical Engineering, vol. 12, no. 1, pp. 1–15, 2020, doi:10.1177/1687814019885294.

[12] D. Rondinelli and M. Berry, "Multimodal transportation, logistics, and the environment: managing interactions in a global economy," European Management Journal, vol. 18, no. 4, pp. 398–410, 2000, doi:10.1016/S0263-2373(00)00029-3.

[13] Thailand Greenhouse Gas Management Organization, Carbon Footprint of Product Guideline, 4th ed. Bangkok: Thailand Greenhouse Gas Management Organization, 2012.

[14] Thailand Greenhouse Gas Management Organization, "Update Emission Factor CFP," 2016. [Online]. Available: http://thaicarbonlabel.tgo.or.th/admin/uploadfiles/emission/ts_822ebb1ed5.pdf.

[15] L. Zadeh, "Optimality and non-scalar-valued performance criteria," IEEE Transactions on Automatic Control, vol. 8, no. 1, pp. 59–60, 1963, doi:10.1109/TAC.1963.1105511.

[16] M. Salukvadze, "On the existence of solutions in problems of optimization under vector-valued criteria," Journal of Optimization Theory and Applications, vol. 13, no. 2, pp. 203–217, 1974, doi:10.1007/BF00935540.

[17] H. Tamaki and E. Nishino, "A Genetic Algorithm Approach to Multi-Objective Scheduling Problems with Regular and Non-Regular Objective Functions," IFAC Proceedings Volumes, vol. 31, no. 20, pp. 283–288, 1998, doi:10.1016/S1474-6670(17)41808-8.

[18] G. Mavrotas, "Effective implementation of the ε-constraint method in Multi-Objective Mathematical Programming problems," Applied Mathematics and Computation, vol. 213, no. 2, pp. 455–465, 2009, doi:10.1016/j.amc.2009.03.037.

[19] G. Chiandussi, M. Codegone, S. Ferrero, and F. E. Varesio, "Comparison of multi-objective optimization methodologies for engineering applications," Computers and Mathematics with Applications, vol. 63, no. 5, pp. 912–942, 2012, doi:10.1016/j.camwa.2011.11.057.

[20] E. Demir et al., "Green intermodal freight transportation : bi-objective modelling and analysis," International Journal of Production Research, vol. 57, no. 19, pp. 6162–6180, 2019, doi:10.1080/00207543.2019.1620363.

Authorized licensed use limited to: UNIVERSITY OF TENNESSEE LIBRARIES. Downloaded on March 20,2024 at 15:26:39 UTC from IEEE Xplore. Restrictions apply.